

ENCN205

Applied Data Analysis for Civil and Natural Systems

LECTURE 7

Cartography Essentials

University of Canterbury | Te Whare Wānanga o Waitaha

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A map is an argument, not a picture.

Maps persuade. Maps simplify. Maps can mislead.

Every design choice — colour, classification, what to include, what to omit — shapes how the reader interprets the data.

As engineers, your maps go to:

Council planners making zoning decisions

Project managers allocating budgets

Communities anxious about hazards

Courts resolving resource consent disputes

Required Map Elements

Every professional map needs:

Title — what the map shows (be specific)

Legend — explains all symbols and colours

Scale bar — indicates distances on the ground

North arrow — orientation (even if obvious)

Data source — provenance and credibility

CRS / projection — what coordinate system

Date — when the data was collected

Missing any of these = unprofessional.

A map without a legend is like a graph without axis labels.

In the exam:

You will be asked to critique a map.

Missing elements = marks lost.

In practice:

Councils will reject submissions with incomplete maps.

Example: Professional Cartometric Map

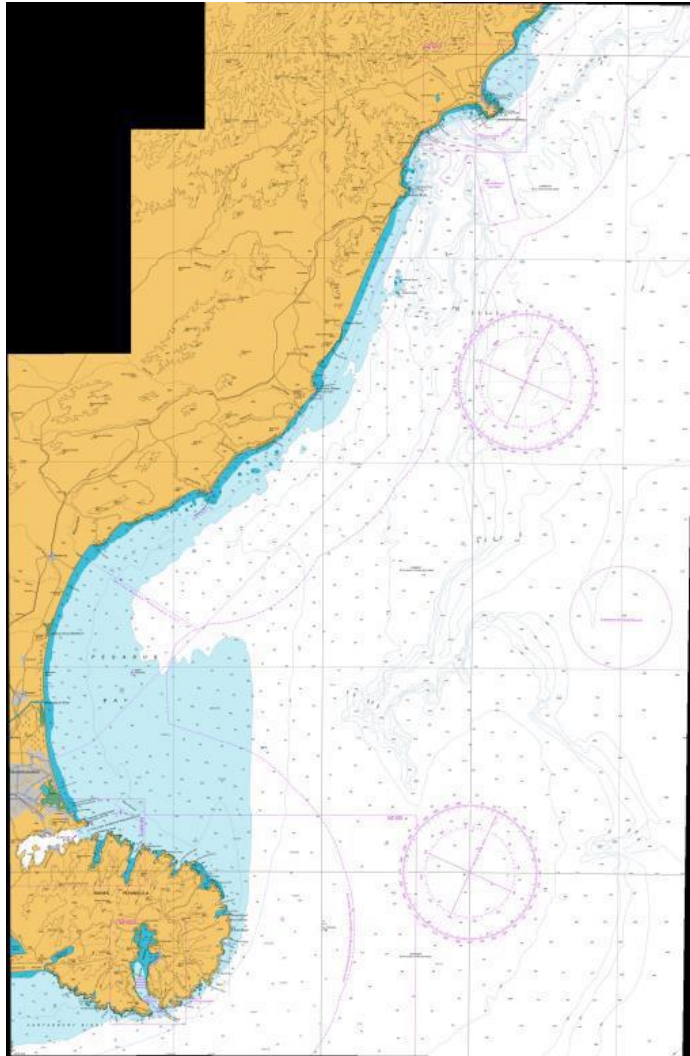


Chart NZ 63 Kaikoura Peninsula to Banks Peninsula
New Zealand Hydrographic Authority

NZ Hydrographic Chart

(LINZ nautical chart)

Spot the required elements:

Title and chart number

Legend (symbolology for depths, features)

Scale bar and compass roses

Coordinate grid (CRS indicated)

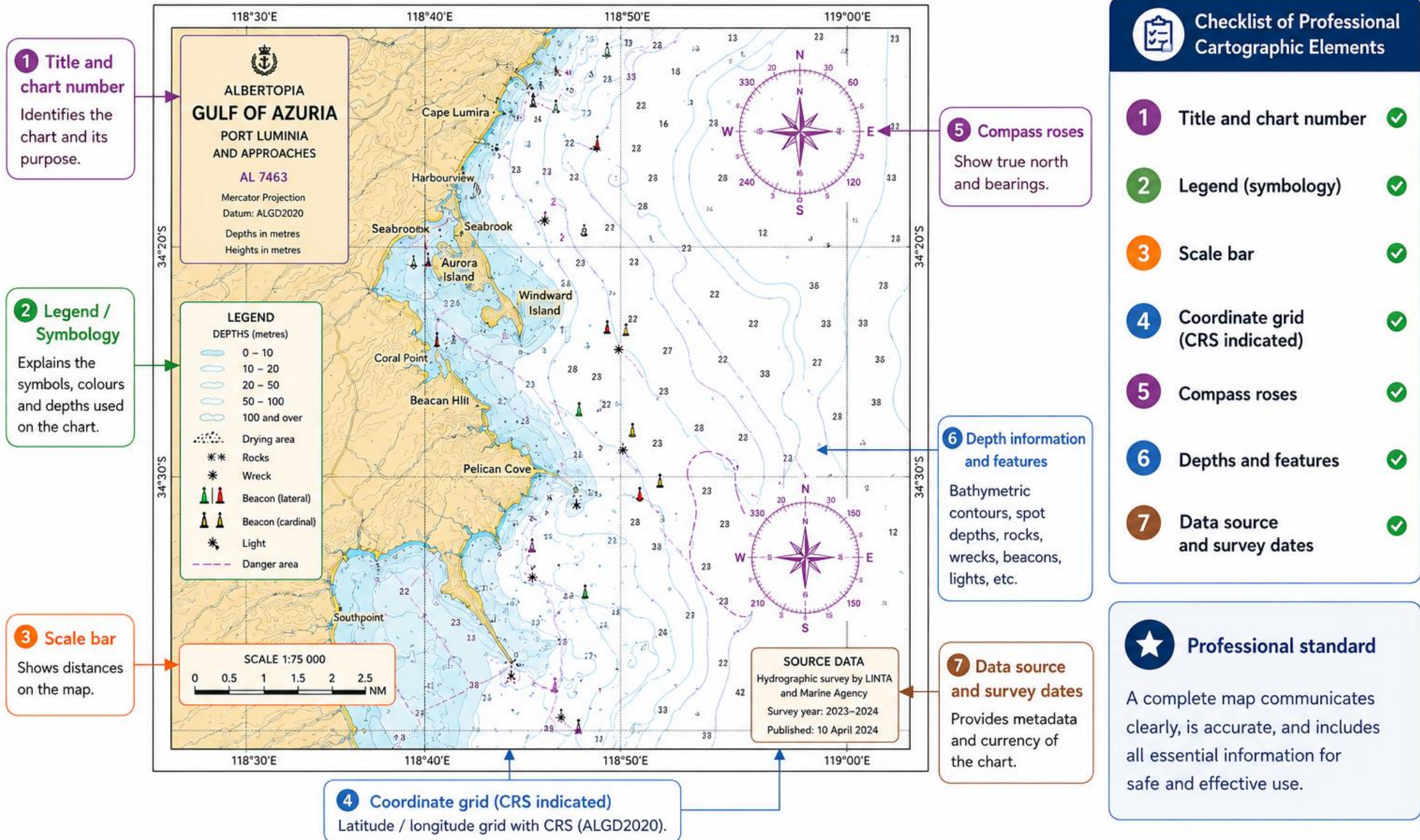
Data source and survey dates

This is professional standard. Your maps should aim for this level of completeness.

Title / chart number: ✓ Chart NZ 63 is identified below the image. **Compass rose:** ✓ Clearly visible. **Coordinate grid:** ✓ Grid lines are visible. **Legend / symbolology:** ⚠ The symbols are visible, but the actual legend/key is not shown in this crop. **Scale bar:** ✗ I can't see one. **CRS / datum information:** ✗ Not visible/readable. **Data source / survey dates:** ✗ Not visible in this crop.

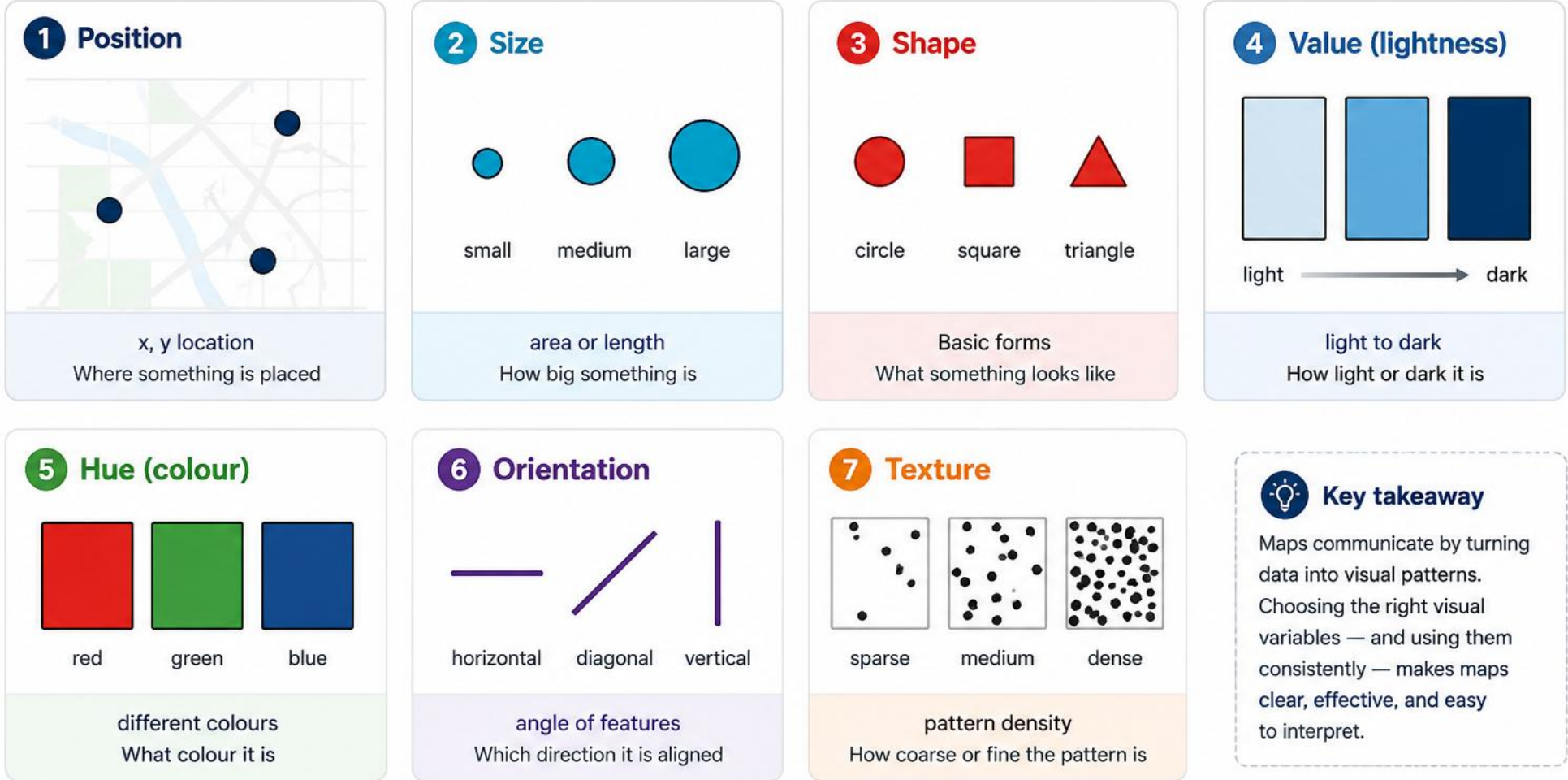
Example: Professional Cartometric Map

Albertopia Hydrographic Chart (LINTA nautical chart example)



Bertin's Visual Variables — The Building Blocks of Map Design

Visual variables control how we encode data on a map. Use them purposefully and consistently.



Good map design principles

- ✓ Match the variable to the data
- ✓ Use only what you need
- ✓ Be consistent
- ✓ Make it easy to read
- ✓ Support the message
- ✓ Test with your users

These are the building blocks of every map symbol. Each variable encodes a different kind of information.

Matching Visual Variables to Data Types

Categorical (nominal) data:

Different types, no order

Use: hue, shape

Land use type (residential, commercial, industrial)

Pipe material (AC, PVC, PE)

DO NOT use: size or value for categories

(implies order that doesn't exist)

Quantitative (ordered) data:

Values with magnitude and order

Use: value (lightness), size

Population density (light = low, dark = high)

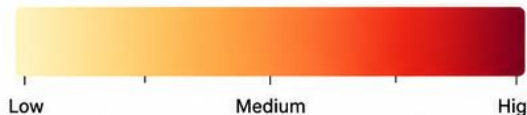
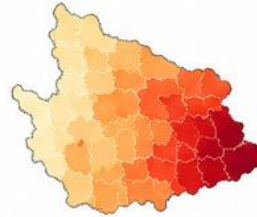
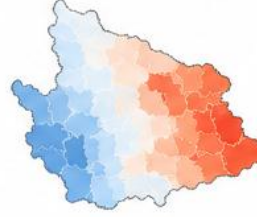
Pipe diameter (small circle = small pipe)

DO NOT use: hue for quantities

(rainbow = no perceptual order)

Three Colour Scheme Types – Choose Based on Your Data

The right colour scheme helps your map communicate the data clearly and effectively.

TYPE & PURPOSE	COLOUR RAMP EXAMPLE	MAP EXAMPLE	WHEN TO USE
 1 SEQUENTIAL For ordered data (light = low, dark = high) Shows an ordered progression from low to high values.	Example colour ramp (yellow–red)  Low Medium High Perceptually uniform, easy to read	Example: Rainfall (mm)  Rainfall (mm) 1,200 900 600 300 0	Use when your data: - Has a natural order - Only goes from low to high - Examples: elevation, population density, rainfall, pipe age  Think: more = darker
 2 DIVERGING For data with a meaningful midpoint (two directions from centre) Highlights values that are above or below a critical midpoint.	Example colour ramp (blue–red)  Negative Zero / Midpoint Positive Use a neutral centre (e.g., white or light grey)	Example: Temperature anomaly (°C)  Anomaly (°C) +4 +2 0 -2 -4	Use when your data: - Has a meaningful centre point (e.g., zero, average, normal) - Shows deviation in two directions - Examples: temperature anomaly, elevation (relative to sea level), change over time  Think: centre = important
 3 QUALITATIVE For categorical data (no inherent order, maximum distinction) Uses distinct colours to separate categories.	Example colour palette (qualitative)  Zone A Zone B Zone C Zone D Zone E Zone F Distinct, high contrast between categories	Example: Land use type  - Residential - Commercial - Industrial - Agriculture - Forest - Water	Use when your data: - Is categorical (groups) - Has no natural order - Examples: land use type, geological unit, pipe material  Think: different = distinct

QUICK GUIDE	GOOD PRACTICE	COMMON MISTAKES TO AVOID								
<table><tr><th>Your data...</th><th>Choose...</th></tr><tr><td>Has order (low → high)</td><td> Sequential</td></tr><tr><td>Has a meaningful midpoint (deviates around centre)</td><td> Diverging</td></tr><tr><td>Is categorical (groups)</td><td> Qualitative</td></tr></table>	Your data...	Choose...	Has order (low → high)	 Sequential	Has a meaningful midpoint (deviates around centre)	 Diverging	Is categorical (groups)	 Qualitative	✔ Use colour purposefully, not decoratively ✔ Ensure colours are perceptually uniform (sequential/diverging) ✔ Check your map in greyscale and colourblind modes ✔ Keep it simple and consistent ✔ Always include a clear legend	✘ Using rainbow colours for ordered data ✘ Using diverging scheme without a meaningful midpoint ✘ Using too many similar colours for categories ✘ Reversing the order (dark = low, light = high) ✘ Missing or unclear legend
Your data...	Choose...									
Has order (low → high)	 Sequential									
Has a meaningful midpoint (deviates around centre)	 Diverging									
Is categorical (groups)	 Qualitative									



Remember

Your colour choices should help readers see the pattern, the message, and the meaning.



Choose the scheme that matches your data structure. ColorBrewer (colorbrewer2.org) is your best friend.

Colour Vision Deficiency

~8% of males and ~0.5% of females have some form of colour vision deficiency.

In a class of 100 engineers, ~4 people cannot distinguish red from green.

Design rules:

Never rely on red-green distinction alone

Use ColorBrewer schemes (all have colourblind-safe options)

Add redundant encoding: colour + pattern, or colour + labels

Test your maps with a CVD simulator (e.g. coblis.myndex.com)

Accessible design is not optional - it's professional practice.

In NZ, the Public Service Act and Building Code require accessible information.

Same Data, Four Classification Methods – Very Different Maps

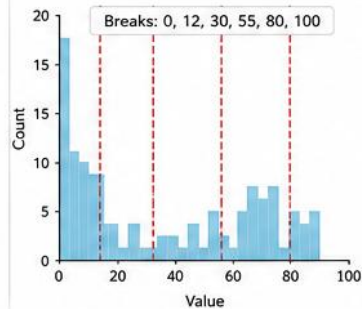
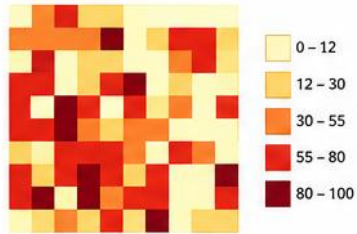
The way we group data into classes changes the story the map tells.



Example data: Values range from 0 to 100 (same data used in all maps)

1 Natural Breaks (Jenks)

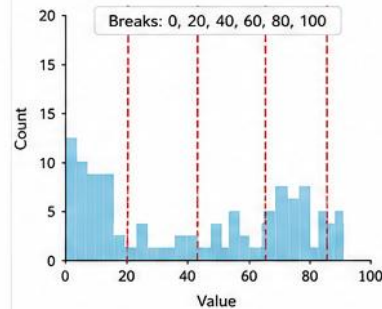
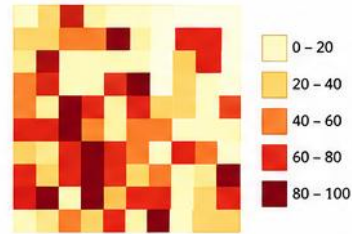
Minimizes variance within classes and maximizes variance between classes.



Best for: Revealing natural groupings and patterns in the data.

2 Equal Interval

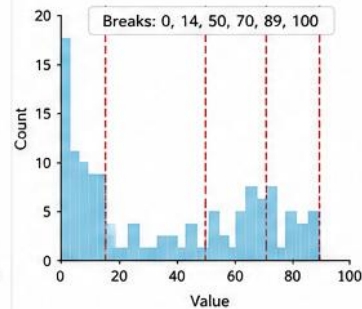
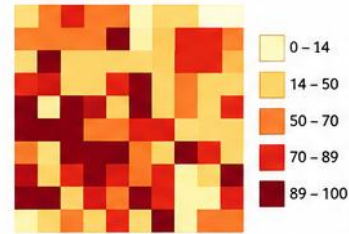
Divides the range into intervals of equal size.



Best for: Data with a known and meaningful scale.

3 Quantile

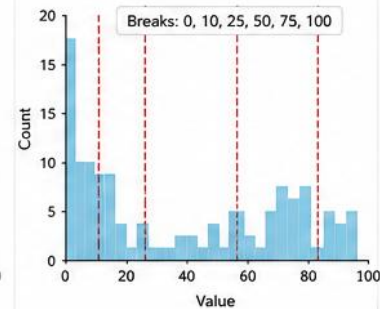
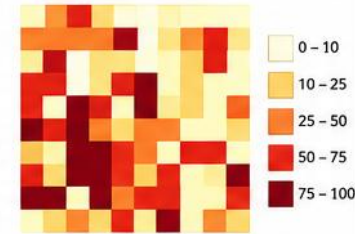
Divides the data into classes with the same number of features.



Best for: Comparing areas equally (e.g., top 20%, middle 20%).

4 Manual (Custom)

User-defined class breaks based on domain knowledge or goals.



Best for: Highlighting meaningful thresholds or policy targets.

Real-world example: Same data, different look

US population density (people per km²)

Equal Interval



Natural Breaks (Jenks)



Quantile



★ Different classification methods can lead to very different interpretations.

How to choose?



Understand your data

What is the distribution? Are there natural groupings or outliers?



Know your goal

Pattern discovery, comparison, or highlighting thresholds?



Choose the method that best matches your objective

There is no one "best" method — the right method depends on your question.

Things to watch out for



Equal Interval can hide important patterns if data are skewed.



Quantile can create classes with very different real-world ranges.



Natural Breaks may change when data change slightly.



Always show your class breaks in the legend!



Remember

Classification is not just a technical step — it's a design choice that shapes the story your map tells.

**Know your data. Know your goal.
Choose your classification wisely.**



Always ask: 'what classification scheme - and why?'

LIVE DEMO

[L07_MapStyling.html](https://aardid.github.io/uc_205_gis_course/L07_MapStyling.html)

Switch classification methods on the same dataset.
See how the map changes — same data, different visual message.

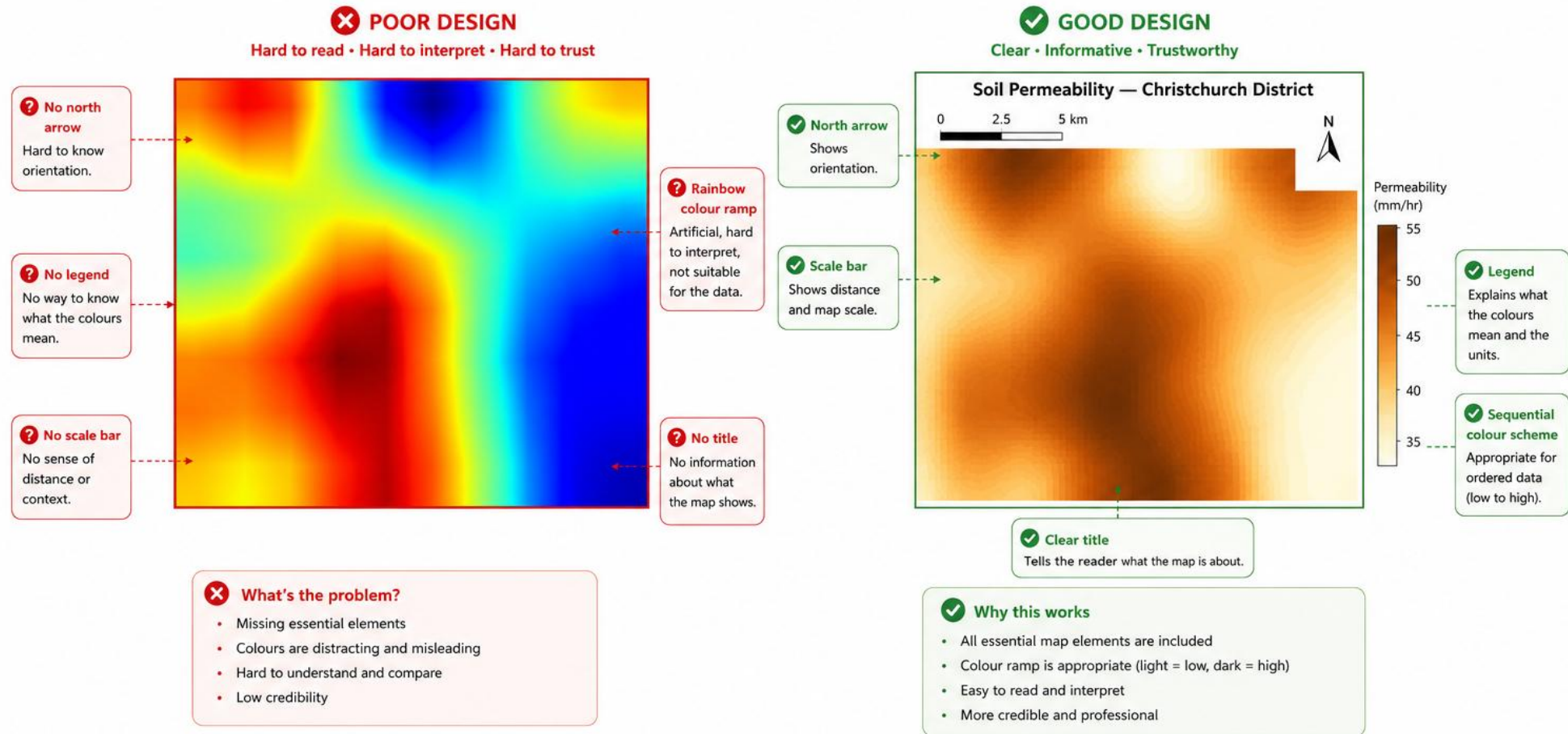
Scan to open the demo on your phone



aardid.github.io/uc_205_gis_course/L07_MapStyling.html

Map Design Matters — The Same Data, Two Very Different Impressions

Good design makes your map easier to read, understand, and trust.



Why map design matters

The same data can tell very different stories depending on how it is displayed. Good design helps your audience understand the message — not fight the map.

Key takeaways

- ✓ Always include essential elements: title, legend, scale bar, north arrow.
- ✓ Use a colour scheme that matches your data and purpose.
- ✓ Keep it simple, consistent, and non-distracting.
- ✓ A well-designed map = clearer communication and better decisions.



Remember

Design is not decoration — it is communication.

Good maps make complex data easy to understand.

Always ask: 'what is this map hiding?'

The Choropleth Trap - Normalise Your Data!

Raw counts in a choropleth = a population density map in disguise.

Example: 'Number of pipe failures by suburb'

Large suburbs have more pipes -> more failures. The map just shows suburb SIZE.

Solution: normalise.

Failures per km of pipe (rate)

Failures per 1000 connections (density)

% of pipes in poor condition (proportion)

```
gdf['rate'] = gdf['failures'] / gdf['pipe_km']
```

Rule: different-sized polygons = MUST normalise.

Imagine two suburbs:

Suburb A: 1,000 pipes, 100 failures

Suburb B: 100 pipes, 30 failures

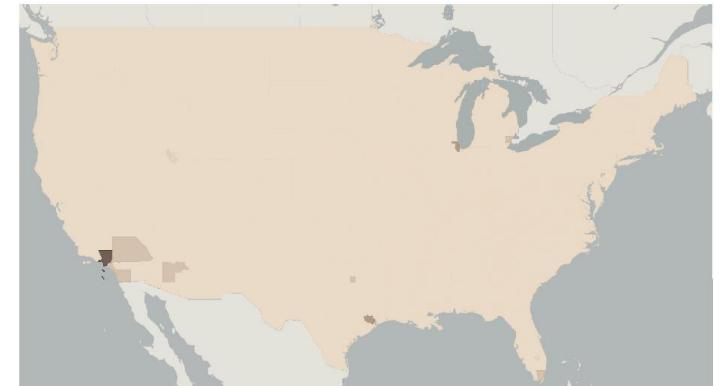
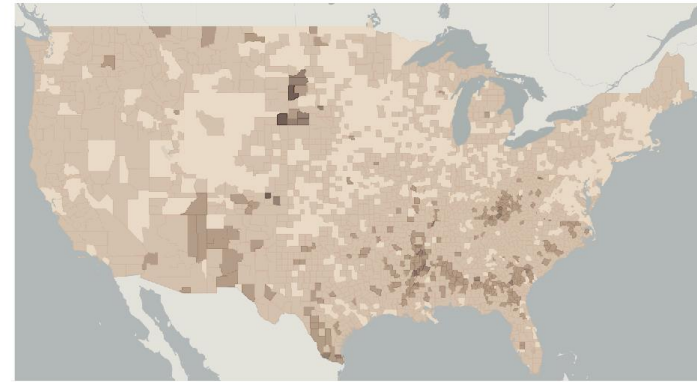
If you map **number of failures**, A looks worse because $100 > 30$.

But calculate the failure rate:

A: $100 / 1,000 = 10\%$

B: $30 / 100 = 30\%$

Now B is clearly in worse condition.



US poverty: percentage (top) vs raw count (bottom) - same data!

Always ask: 'counts or rates?'

How Maps Deceive

1. Cherry-picking classification breaks

Choose breaks that make your story more dramatic (or less).

2. Misleading colour schemes

Red for low values makes everywhere look 'alarming'.

3. Truncated legends

Legend range 50-100 when data is 0-100 — hides the low end.

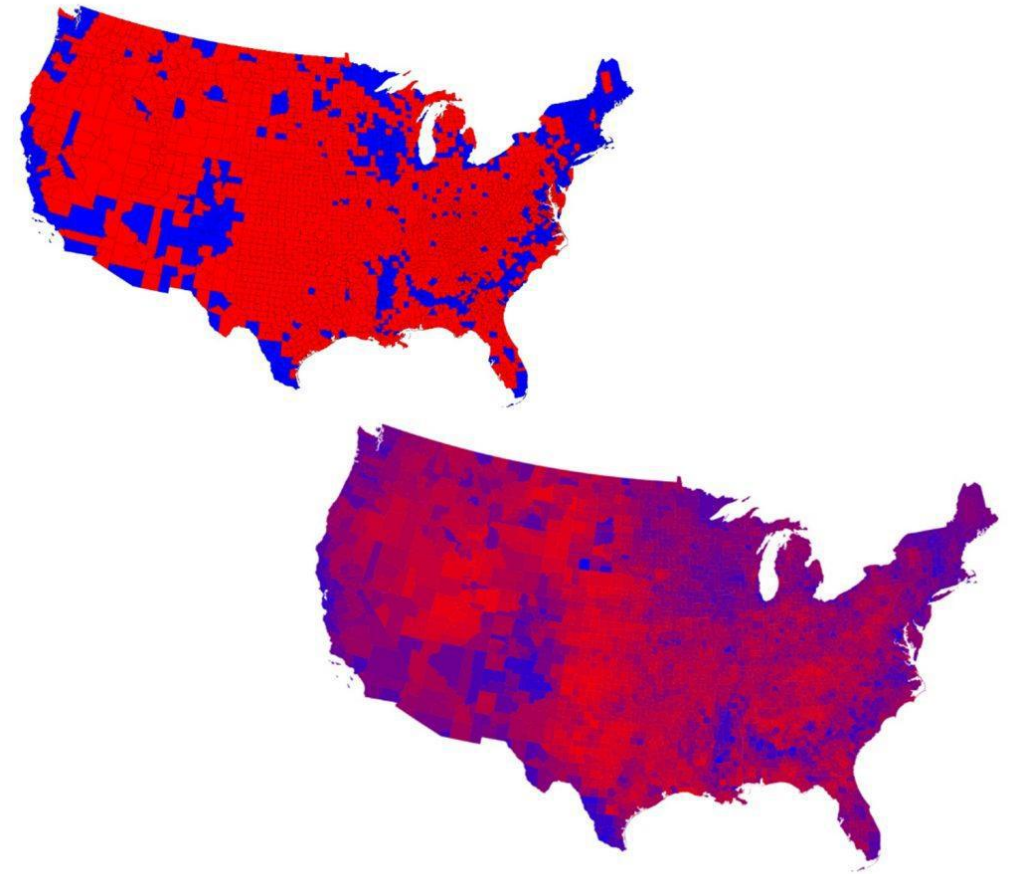
4. Omitting inconvenient data

Excluding outlier regions that don't fit the narrative.

Ethical obligation: represent data honestly.

Maps don't just display data - design choices can strongly influence the story people see

US Presidential Election Results by County (2012)



US 2012 election: county-level (top) vs blended representation (bottom)

Map Design for Engineering Reports

What clients and councils expect:

- All seven map elements present
- Professional colour scheme (no rainbow)
- Clear, readable legend with units
- Appropriate classification with method stated
- Company/agency branding where required
- Print-friendly (readable at A3 or A4)

Practical tips:

- White or light grey background
- Max 5-7 classes for choropleths
- Consistent symbology across a report
- Include a locator/inset map if needed
- Label key features (roads, landmarks)
- Export at 300 dpi for print, 150 for screen

Practical Tools for Colour Selection

ColorBrewer (colorbrewer2.org)

Interactive web tool for choosing map colours

Filter by: colourblind-safe, print-friendly

Export hex codes for any software

Top recommendations:

Sequential: 'YlOrRd', 'Blues', 'Greens'

Diverging: 'RdBu', 'BrBG', 'PiYG'

Qualitative: 'Set2', 'Paired', 'Dark2'

In matplotlib / GeoPandas:

```
gdf.plot(column='value',  
         cmap='YlOrRd',  
         scheme='NaturalBreaks',  
         k=5,  
         legend=True)
```

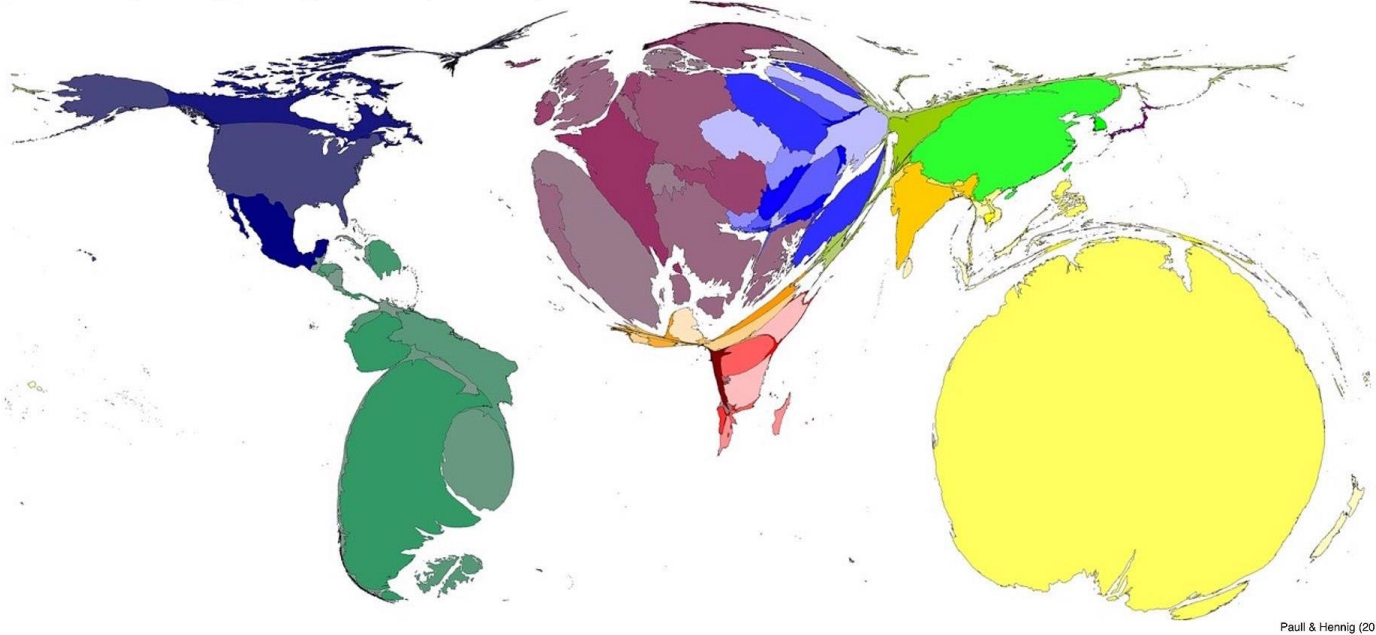
scheme options:

'NaturalBreaks', 'EqualInterval',
'Quantiles', 'UserDefined'

Requires: pip install mapclassify

Example: When Maps Distort Reality

World Map of Organic Agriculture (hectares)



Cartogram: Organic Agriculture

Country area = hectares of organic farmland

What's happening:

Geography is distorted to encode data
Australia dominates (huge organic area)
Some countries nearly disappear

Discussion:

Is this misleading or informative?
What context is missing?
What would a choropleth show?

Cartograms sacrifice familiar geography for data impact.

Today's Takeaways

1. Every professional map needs: title, legend, scale bar, north arrow, source, CRS, date
2. Match visual variables to data type: hue for categories, value/size for quantities
3. Classification method changes the story — always state which method you used
4. Normalise data before mapping in choropleths (rates, not raw counts)
5. Design for accessibility: colourblind-safe palettes, redundant encoding

**You're mapping pipe condition ratings (1-5) for a council report.
What colour scheme and classification would you use?**

Think about:

Is this data categorical or quantitative? (It's ordered: 1=good, 5=critical)

Sequential or diverging scheme?

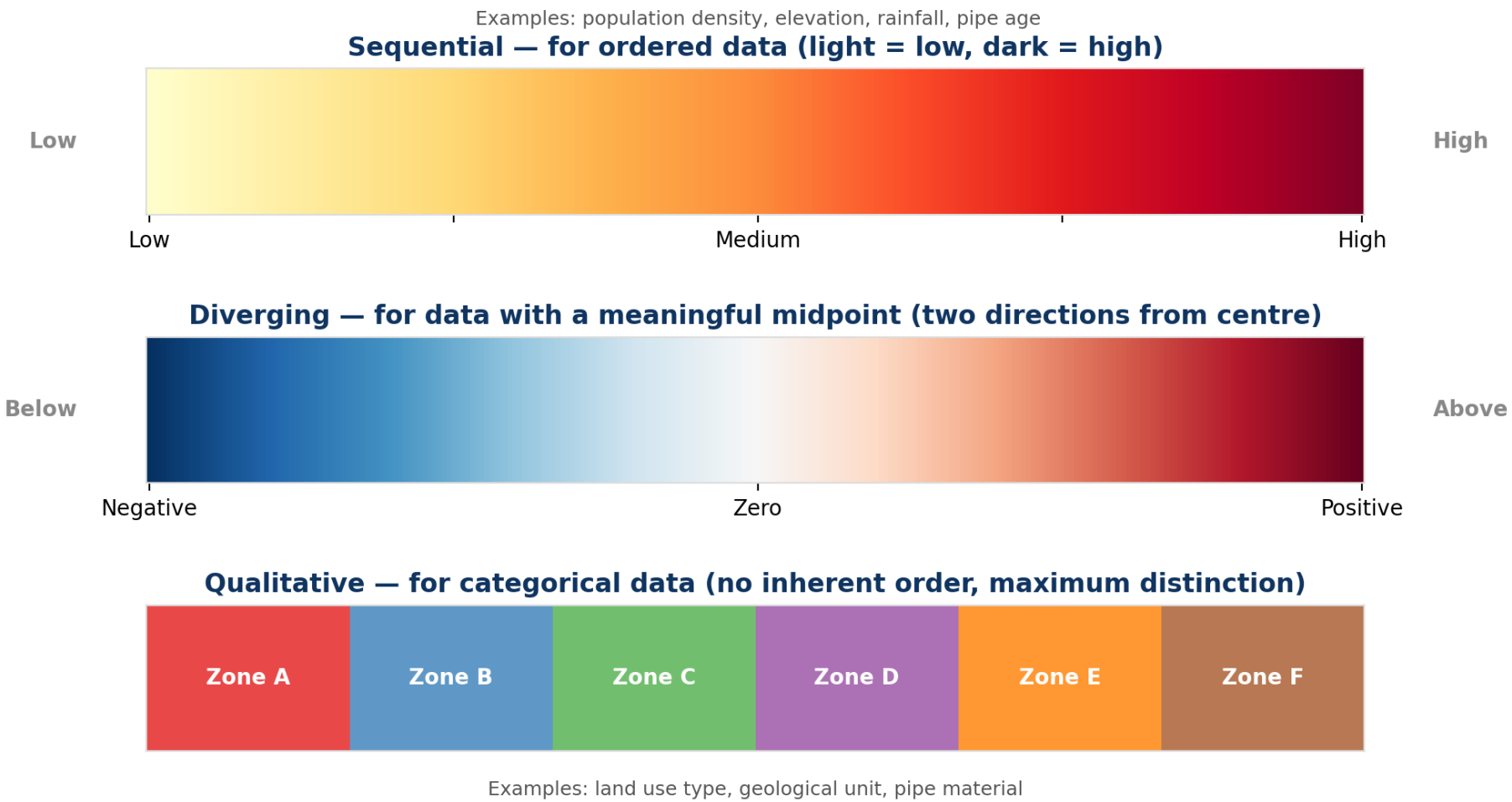
What classification? (Hint: the ratings ARE the classes)

Should you normalise? (Per suburb? Per km of pipe?)

Bonus: how would you handle a suburb with only 2 pipes vs one with 200?

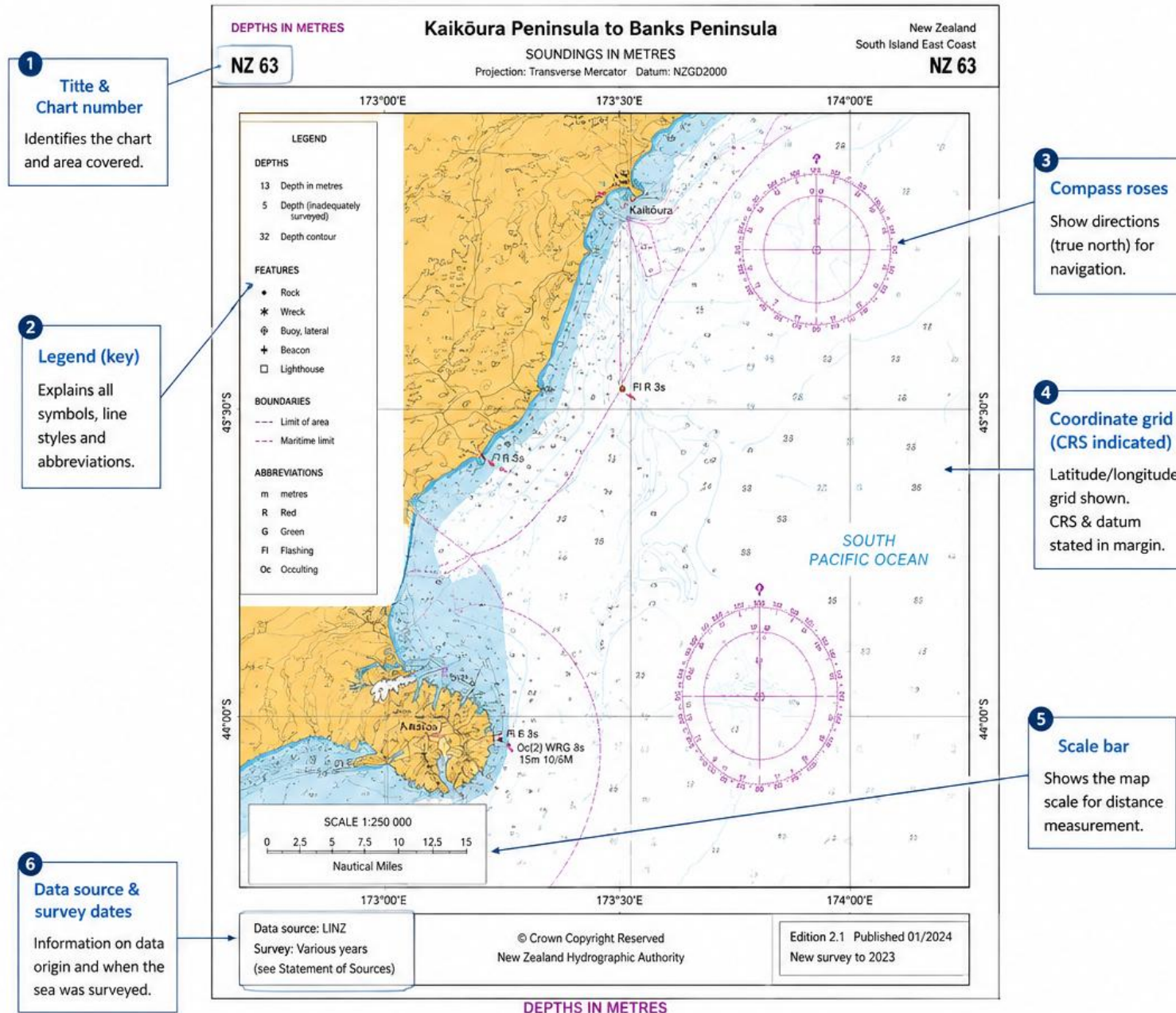
Colour Schemes

Three Colour Scheme Types — Choose Based on Your Data



Choose the scheme that matches your data structure. ColorBrewer (colorbrewer2.org) is your best friend.

Example: Professional Cartometric Map



NZ Hydrographic Chart
(LINZ nautical chart)

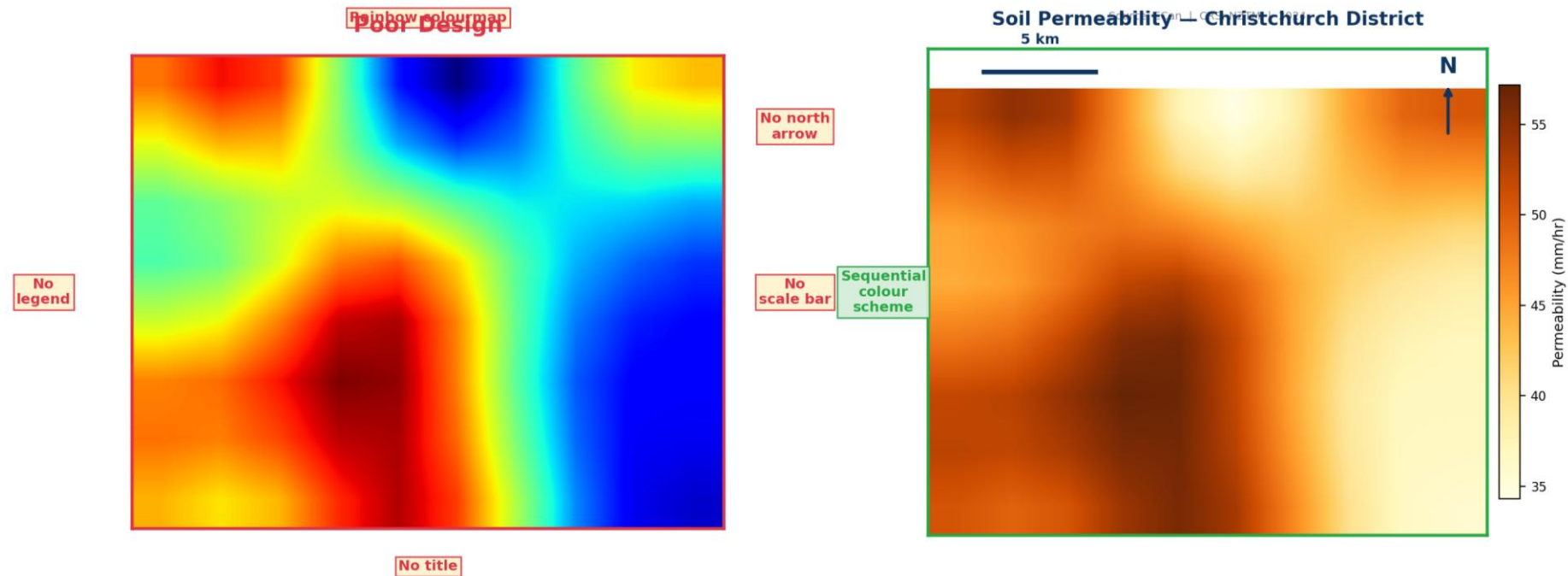
Professional cartographic elements:

- 1 Title & chart number**
Identifies the map and area.
- 2 Legend (symbology)**
Explains all symbols, lines, colours and abbreviations.
- 3 Compass roses**
Indicate directions (true north).
- 4 Coordinate grid (CRS indicated)**
Lat/long grid shown; projection and datum stated in the margin.
- 5 Scale bar**
Allows accurate distance measurement.
- 6 Data source & survey dates**
Shows data origin and when surveys were conducted.

Professional standard

A professional map communicates:
where you are, what it shows,
how to read it, how accurate it is,
and where the data came from.

Critique: Good vs Bad Map Design



The left map has the same data as the right — but communicates almost nothing useful.

Always ask: 'what is this map hiding?'

Classification Choice Matters

Same data, very different stories:

Natural Breaks: emphasises clusters in the data

Equal Interval: emphasises the range

Quantile: forces visual equality

Manual: enforces engineering thresholds

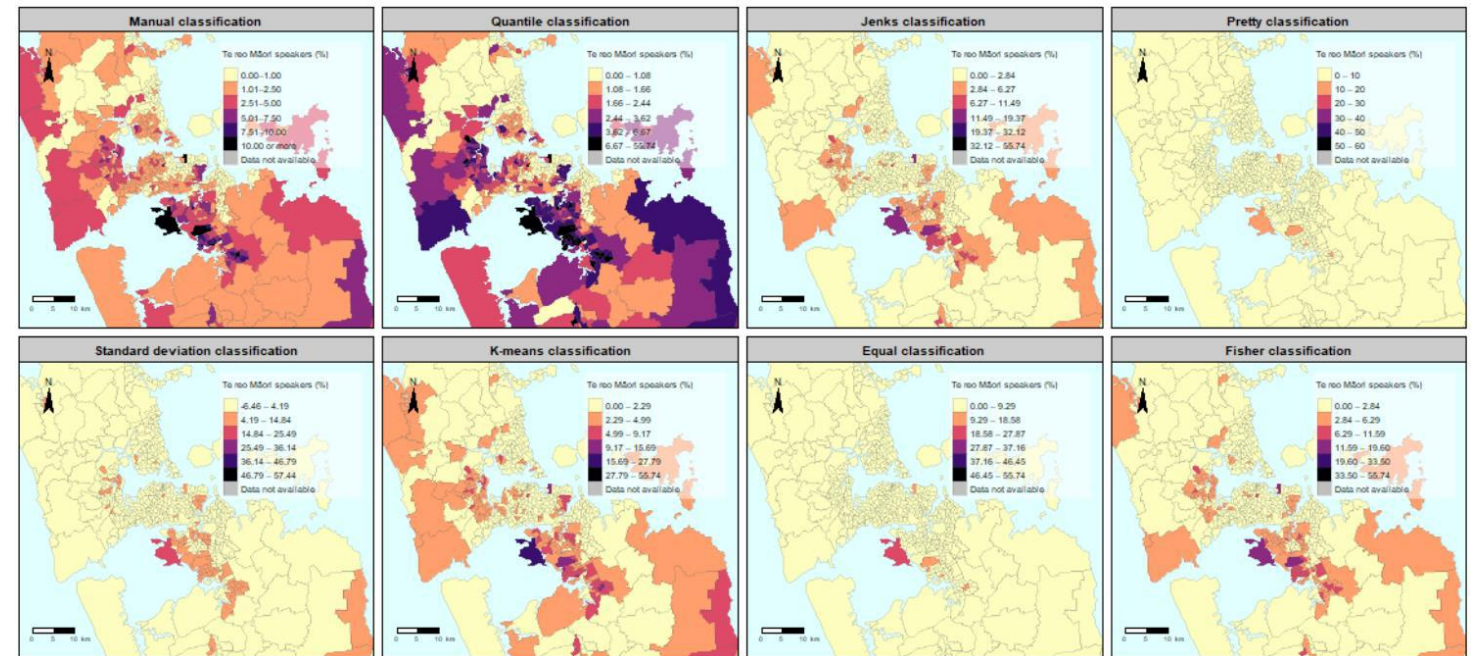
Std Dev, K-means, Pretty: statistical variants

Recommendation:

Start with Natural Breaks for exploration.

Use Manual when standards define breaks.

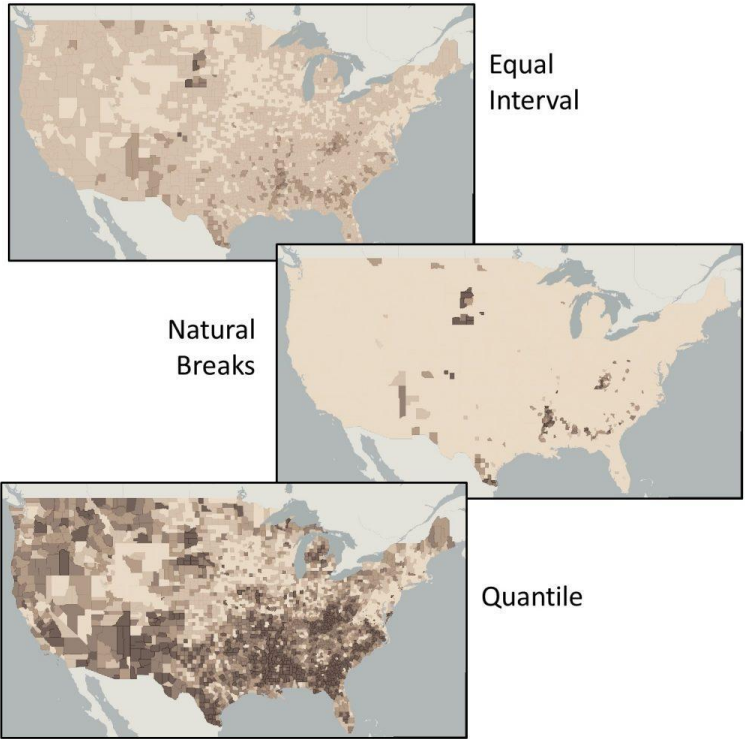
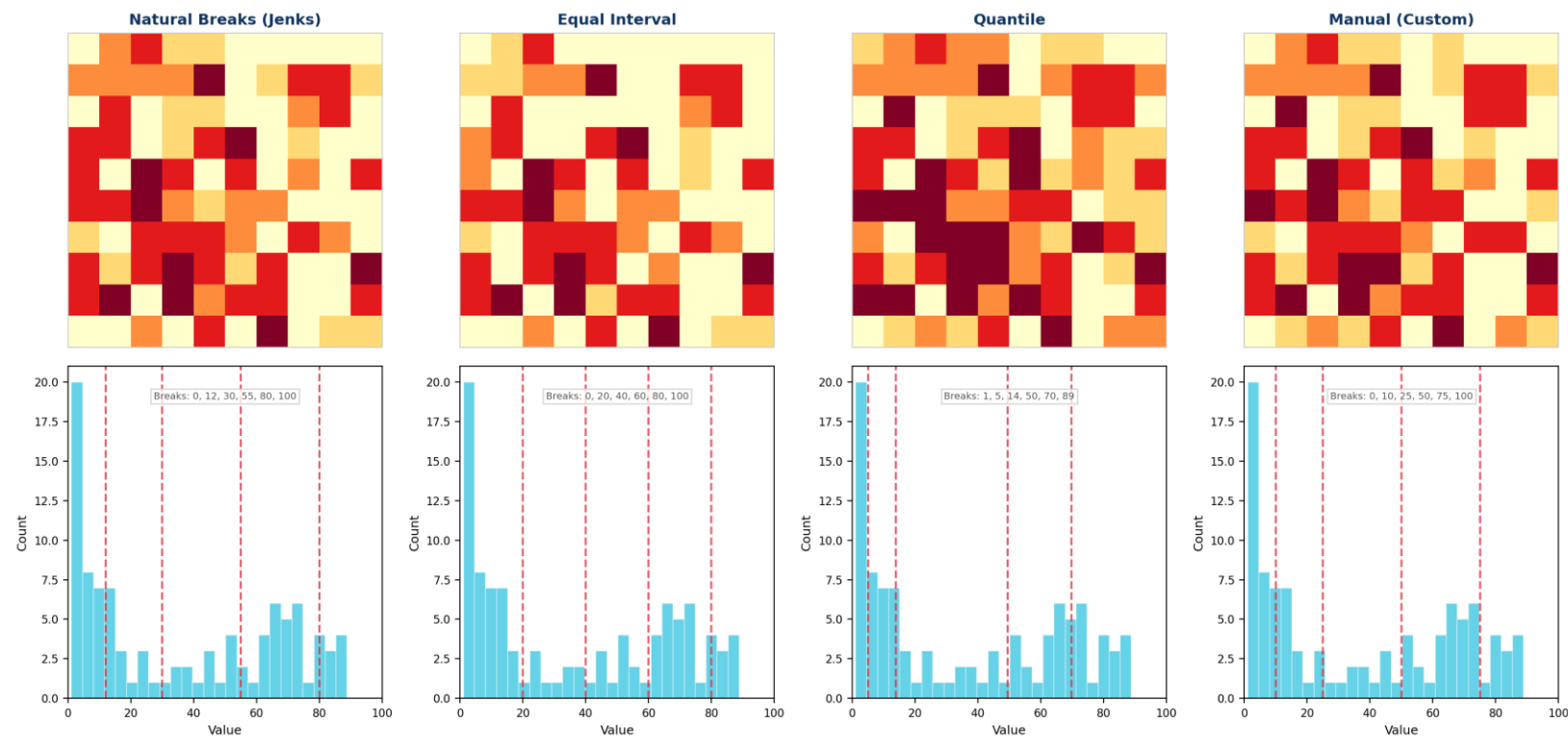
Always state the method in your legend.



Seven classification methods applied to the same dataset

Data Classification Methods

Same Data, Four Classification Methods — Very Different Maps



Same US poverty data,
3 classification methods

Always ask: 'what classification scheme - and why?'